

A PROJECT TITLE

ON

Explainable Hybrid Machine Learning Framework for Predictive Maintenance

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

In

A.Y.: 2026-27

By

Name V. Navadeep

Roll 2520030019

Name K. Shankar

Roll 2520030063

Under the Esteemed guidance of

Nirmala Jyothi Narisetty

CSE



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



DEPARTMENT OF _____

Name of Title: Explainable Hybrid Machine Learning Framework for Predictive Maintenance

1. Abstract:

The **Explainable Hybrid Machine Learning Framework for Predictive Maintenance** is an Industrial Internet of Things (IIoT)-based system designed to predict potential machine failures before they occur. In modern industrial environments, unexpected equipment failures can lead to production downtime, increased maintenance costs, equipment damage, and reduced operational efficiency. Traditional maintenance approaches, such as reactive and scheduled maintenance, may not effectively identify the actual condition of equipment or provide timely warnings. To address these challenges, the proposed framework uses machine learning techniques to analyze data collected from industrial machine sensors, including temperature, vibration, pressure, rotational speed, operating hours, and other relevant parameters.

The proposed system follows a hybrid machine learning approach in which multiple machine learning models are utilized to improve prediction performance and reliability. The system performs data preprocessing, feature selection, model training, and evaluation to identify patterns associated with normal and abnormal machine conditions. Based on these patterns, the framework predicts whether a machine is likely to experience a failure, allowing maintenance activities to be planned before a critical breakdown occurs.

A key feature of the proposed framework is the integration of **Explainable Artificial Intelligence (XAI)** techniques. Instead of providing only a failure prediction, the system explains the major factors that influenced the prediction. Techniques such as SHAP or LIME can be used to identify the contribution of individual sensor parameters to the model's decision. This improves transparency, builds user confidence, and helps maintenance personnel understand the condition of industrial equipment.

The proposed framework aims to reduce unplanned downtime, minimize unnecessary maintenance, lower operational costs, and improve machine reliability. By combining IIoT sensor data, hybrid machine learning, and explainable AI, the system provides a practical and intelligent solution for predictive maintenance in industrial environments. The framework can ultimately support data-driven maintenance decisions and improve overall industrial productivity and equipment performance.

2. INTRODUCTION:

Machine failures in industries can cause unexpected downtime, high maintenance costs, and production losses. Our proposed **Explainable Hybrid Machine Learning Framework for Predictive Maintenance** uses **Industrial IoT (IIoT)** sensors to continuously collect machine data such as temperature, vibration, pressure, and speed. Machine Learning models analyze this data to identify unusual patterns and predict possible failures before they happen. The hybrid approach combines different models to improve prediction accuracy, while **Explainable AI (XAI)** shows the main factors responsible for each prediction. This makes the system easier for maintenance teams to understand and trust. By providing early warnings and clear explanations, the proposed system helps industries plan maintenance at the right time, reduce unexpected breakdowns and costs, improve machine reliability, and maintain efficient production.

3. Software Requirements:

- **Operating System:** Windows 10/11 or Linux
- **Programming Language:** Python 3.x
- **Development Environment:** Jupyter Notebook / Google Colab / VS Code
- **Data Processing:** Pandas, NumPy
- **Data Visualization:** Matplotlib, Seaborn
- **Machine Learning:** Scikit-learn
- **Explainable AI:** SHAP and/or LIME
- **Dataset Format:** CSV

The system will use Python-based machine learning libraries for data preprocessing, model training, prediction, evaluation, and explainability. Sensor datasets stored in CSV format will be used to train and test the predictive maintenance models.

Signature of the guide

Course Instructor